

Can we put a price on climate impacts?

Heat-related mortality and valuing mortality risk

ECON-4210

Economics of Climate Change

Class 12

Prof. Casey Wichman

Today's class

1. Questions?
2. Attendance
3. Housekeeping
 1. Problem set 2 is due Friday, 10/16, 11:59 PM.
 2. Exam 2 is Monday, 10/19 (classes 9–13). Review on Wednesday, 10/14.
4. Ghost Emissions
5. Heat and mortality
6. Valuing mortality risk
7. For next class...

Where are we?

1. So far, we've focused on climate damages with prices:
 - Prof. Wichman's electricity bill
 - GDP (Burke, Hsiang, & Miguel)
 - Corn and soybean yields (Schlenker & Roberts)
2. Today: deaths.
 - How does the death rate respond to temperature?
 - How do we put a dollar value on that?
3. On Monday, we'll combine all the pieces into the **social cost of carbon**.

**How does an
additional hot day
affect mortality?**

What do Carleton et al. (2022) do?

1. **Data:** annual death rates by age for counties or provinces in 40 countries (38% of the world's population), matched to daily temperature.
2. **Approx. same design as my electricity bill:** regress death rate on daily temperature, with location and year fixed effects, by age group (<5, 5–64, >64).
3. The shape of the response depends on a region's income and usual climate. That gives a predicted curve for 24,378 regions worldwide.
4. **Project to 2100:** 33 climate models, two emissions paths (RCP4.5, RCP8.5), projected income growth.
5. Adaptation flattens the curve. Inferred from hotter and richer places that have already adapted (where MC = MB).
6. No migration, but population exposed to extreme heat changes based on anticipated population growth.

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JOURNAL ARTICLE

Valuing the Global Mortality Consequences of Climate Change Accounting for Adaptation Costs and Benefits*

Tamma Carleton, Amir Jina, Michael Delgado, Michael Greenstone, Trevor Houser, Solomon Hsiang, Andrew Hultgren, Robert E Kopp, Kelly E McCusker, Ishan Nath ... [Show more](#)

The Quarterly Journal of Economics, Volume 137, Issue 4, November 2022, Pages 2037–2105, <https://doi.org/10.1093/qje/qjac020>
Published: 21 April 2022

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Abstract

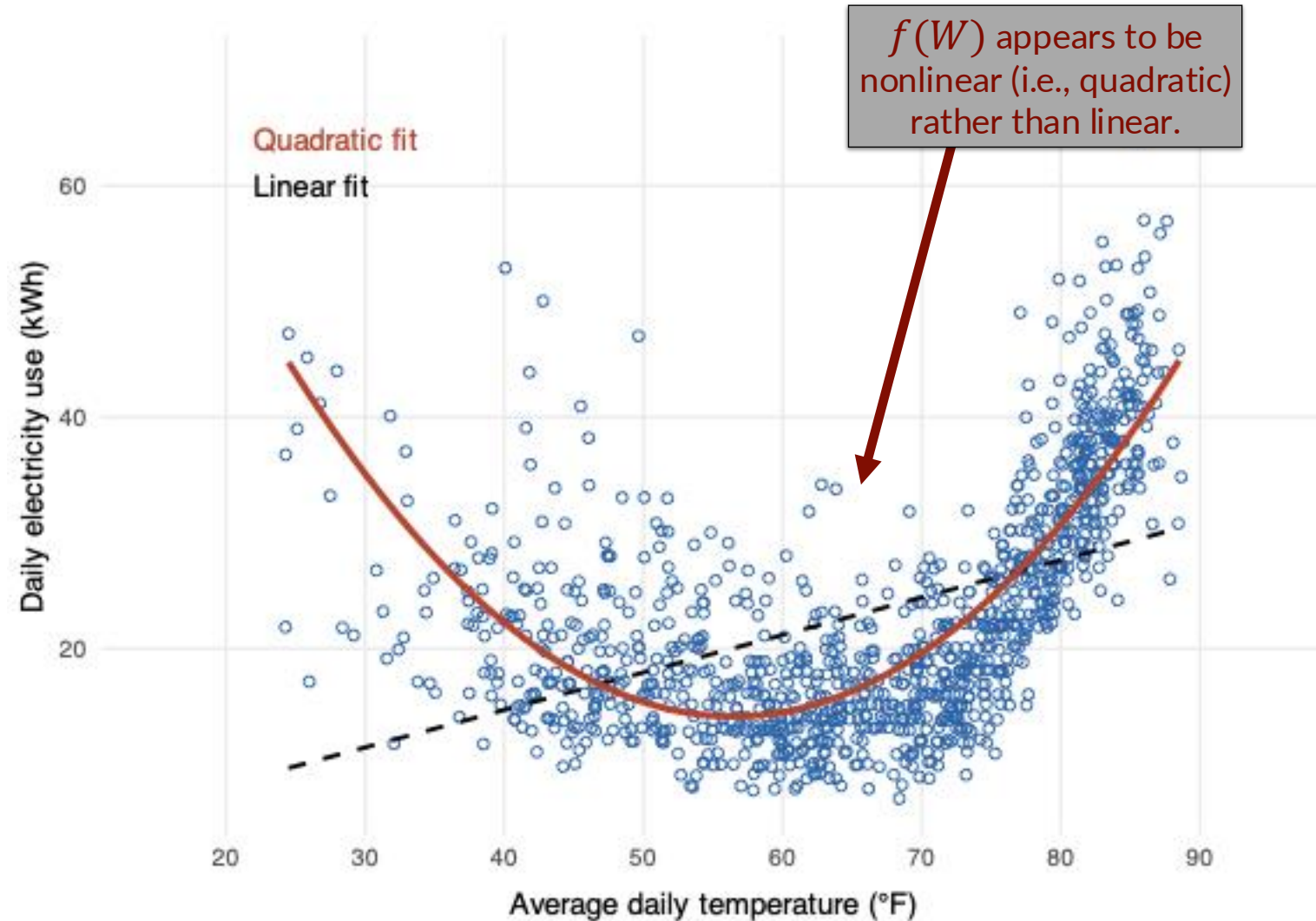
Using 40 countries' subnational data, we estimate age-specific mortality-temperature relationships and extrapolate them to countries without data today and into a future with climate change. We uncover a U-shaped relationship where extreme cold and hot temperatures increase mortality rates, especially for the elderly. Critically, this relationship is flattened by higher incomes and adaptation to local climate. Using a revealed-preference approach to recover unobserved adaptation costs, we estimate that the mean global increase in mortality risk due to climate change, accounting for adaptation benefits and costs, is valued at roughly 3.2% of global GDP in 2100 under a high-emissions scenario. Notably, today's cold locations are projected to benefit, while today's poor and hot locations have large projected damages. Finally, our central estimates indicate that the release of an additional ton of CO₂ today will cause mortality-related damages of \$36.6 under a high-emissions scenario, with an interquartile range accounting for both econometric and climate uncertainty of [−\$7.8, \$73.0]. These empirically grounded estimates exceed the previous literature's estimates by an order of magnitude.

JEL: [H23 - Externalities; Redistributive Effects; Environmental Taxes and Subsidies](#), [H41 - Public Goods](#), [I14 - Health and Inequality](#), [Q51 - Valuation of Environmental Effects](#), [Q54 - Climate: Natural Disasters; Global Warming](#)

Issue Section: [Article](#)

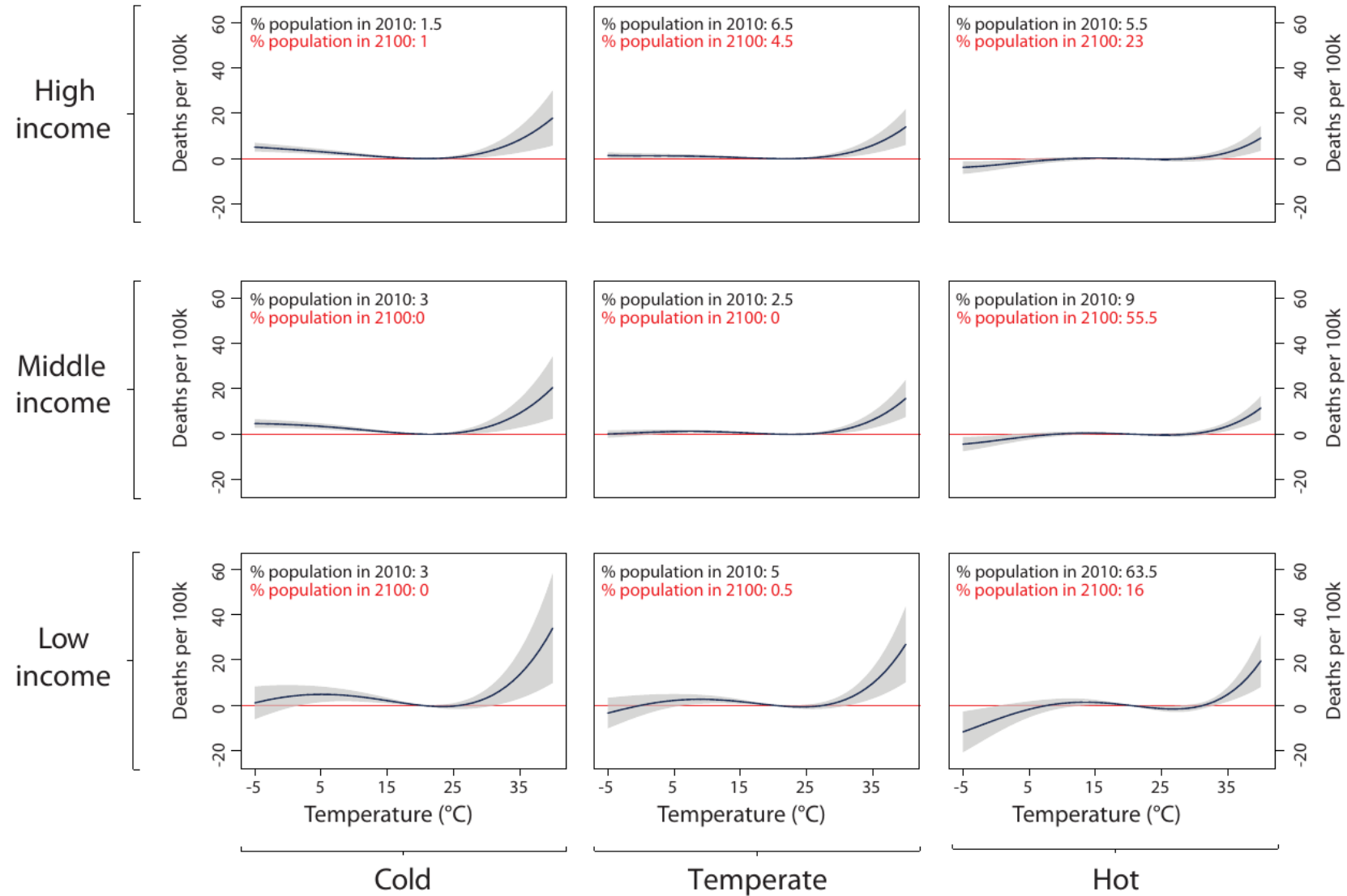
Same idea as my electricity bill

- **Question:** how does my electricity use change with weather?
- **Data:**
 - Daily kWh usage since Sep 2023 (downloaded from Georgia Power)
 - Daily average outdoor temperatures.
- Obviously, electricity use is seasonal and responds to weather.
- I use more AC when it's hot out. I use more lights when it's dark out.



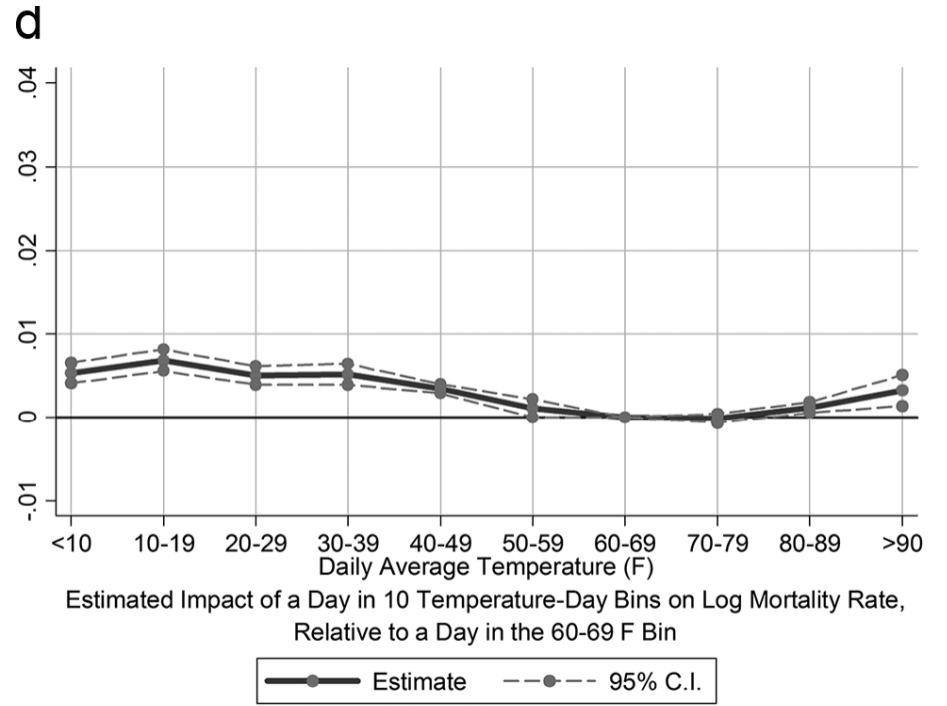
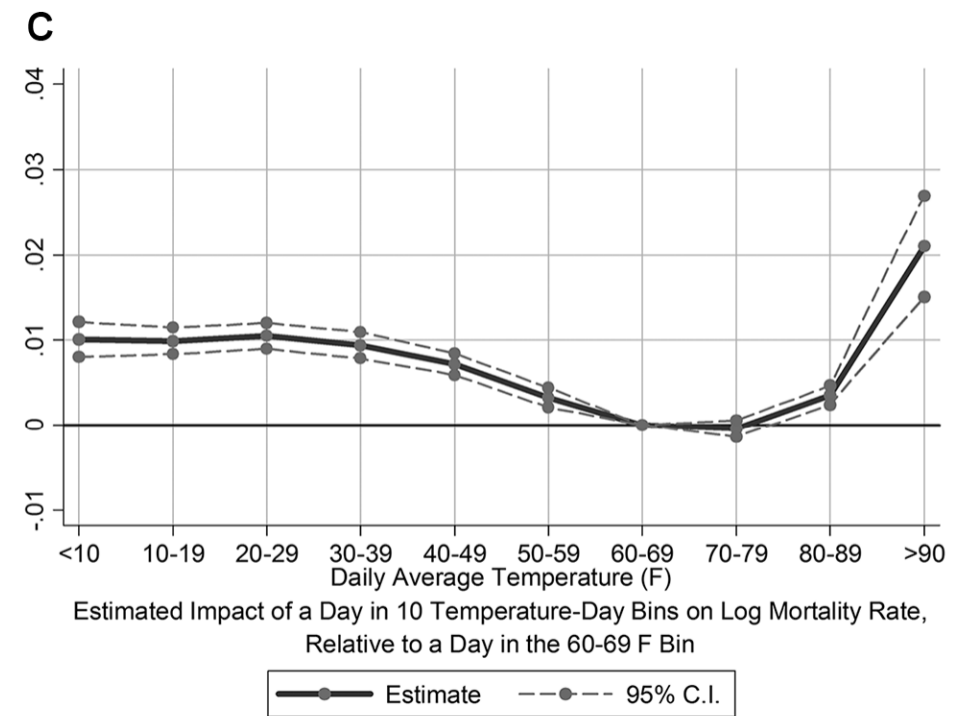
Mortality-temperature response functions

- Deaths rise at both ends relative to a mild ($\sim 20^{\circ}\text{C}$ / 68°F) day.
- People over 64 drive almost all of the effect:
 - +4.7 deaths per 100k on a 35°C (95°F) day
 - +3.4 deaths per 100k on a -5°C (23°F) day
- Cold regions have steeper hot tails.
- High-income regions are nearly flat on the hot side.
 - Why?
- Low-income regions have the steepest hot tail.
- In 2010, 63.5% of the world's population lived in the low-income, hot cell.
 - In 2100, 55.5% of pop is in middle-income/hot



Heat-related mortality has declined over time in the US

- Different paper: Barreca et al., 2016.
- Mortality rates from US states, 1931–2004
- Effect of one day in each temperature bin on the monthly death rate, relative to a 60–69°F day.
 - Top panel: 1931–59
 - Bottom panel: 1960–2004
- A day above 90°F raised monthly mortality by:
 - 2.2% before 1960, and
 - 0.3% after 1960.
 - An 80% decline...
- What caused this dramatic decline in heat-related mortality in the US?



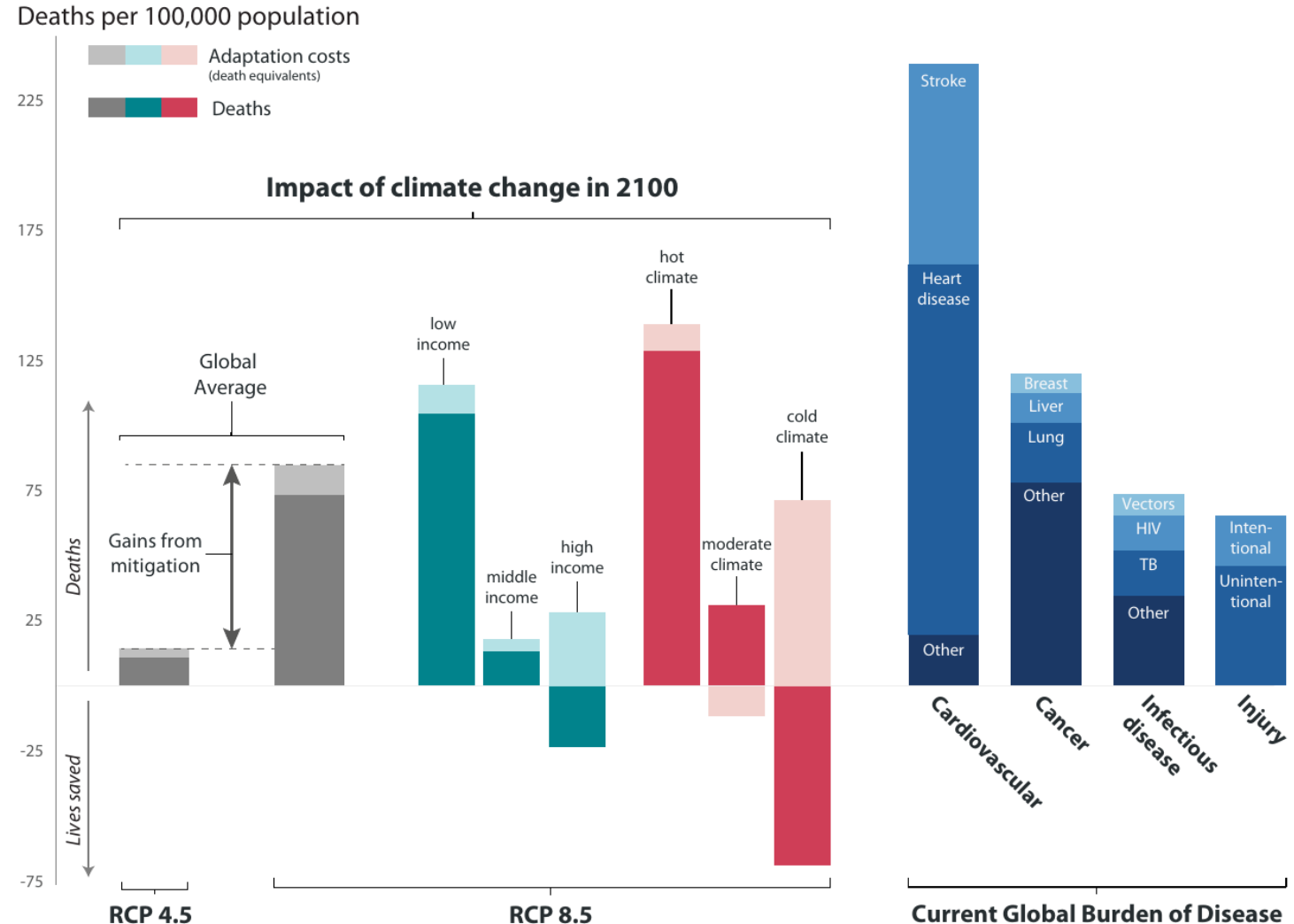
Source: Barreca et al.
(2016), JPE, Fig. 2 panels
c and d.

Adaptation is costly

1. Flatter curves come from spending on adaptation: air conditioners, electricity demand, better housing/insulation, staying inside.
2. Carleton et al. (2022) count both mortality costs + adaptation costs:
 - Deaths that still happen after people adapt (but fewer)
 - Money spent avoiding heat-related mortality (= “adaptation costs”)
3. Climate projections allow income to increase, which flattens curves on its own.
4. Poorer countries have the steepest curves and the least amount of resources to adapt.

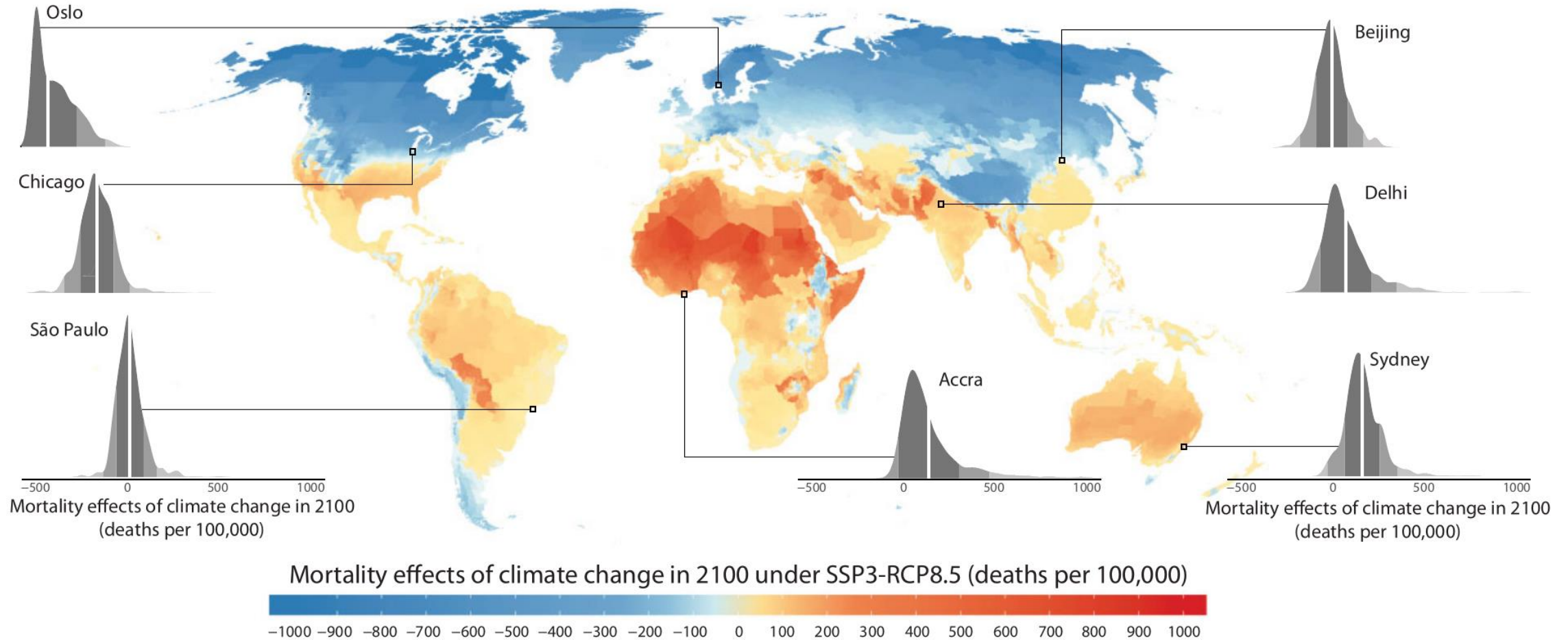
How important is heat-related mortality in 2100?

- Global average under high emissions (RCP8.5), after income growth and adaptation: 73 deaths per 100,000 per year.
 - Without income growth or adaptation: 221.
 - Moderate emissions (RCP4.5): about 11.
 - No migration in any of these numbers.
- For scale: US traffic deaths are ~12 per 100,000.
- Lighter bars are adaptation costs, converted to death equivalents.
- Climate-induced heat-related mortality is ~similar magnitude as today's leading causes of death.



Source: Carleton et al. (2022), QJE, Fig. IX.

Where do heat-related deaths occur?



- Deaths per 100,000 in 2100, SSP3-RCP8.5.
- Hot, poor regions lose. Accra: about +160 per 100,000 per year.
- Cold, rich regions gain from fewer cold days. Berlin: about -150 per 100,000.
- The global average of 73 deaths per 100k masks differences across geographies.
- People are held in place. With migration out of the red regions, both the losses and the gains would shrink.

Source: Carleton et al. (2022), QJE, Fig. IV.

**How do we put a
dollar value on a
mortality?**

The value of a statistical life

a.k.a. the value of mortality risk reduction

1. Economists/policymakers do not value a particular person's life. We value small reductions in the risk of mortality.
2. The value of a statistical life (VSL) is the total amount people are willing to pay for risk reductions that add up to one fewer expected death.
3. Grab 100,000 strangers (Bobby Dodd Stadium holds about 52,000).
 - Ask each one how much they would pay to cut their chance of dying this year by 1 in 100,000.
 - If everyone says \$100, then $VSL = \$100 \times 100,000 = \10 million .
 - How small is 1 in 100,000?
 - Approximately the risk of dying from general anesthesia during surgery (0.5-1 in 100,000 overall, far lower for healthy patients), or about one month of driving in the US (~12 deaths per 100,000 per year).
4. Tends to be the largest benefit category in analyses of environmental regulation.

Where does the VSL come from?

1. Revealed preference: what people actually trade for safety.
 - Wage premiums in risky jobs (roofers, loggers, fishermen vs. similar workers in safe jobs)
 - Purchases: safer cars, smoke detectors, bike helmets
2. Stated preference: surveys that ask people what they would pay for a risk reduction.
3. US EPA's current estimate: ~\$11.7 million (2025 dollars).
 1. Carleton et al. use EPA's \$10.95 million (2019 dollars).

One VSL for everyone?

1. In US regulation: yes.
 - One VSL regardless of age, income, or location.
2. However, evidence says willingness to pay for risk reductions rises with income and changes with age.
3. Climate deaths from Carleton et al. occur mostly in poor countries. How do we transfer a US VSL there?
 - Scale by country-specific median income (Carleton et al.): a death in a poor country counts for fewer dollars.
 - If Carleton et al. use one global VSL: global damages rise (by a lot).
 - Is this ethically defensible?
4. Can we actually put a price on life?
 1. Speed limits, building codes, etc., already does. The VSL makes the tradeoff explicit.

From deaths to dollars per ton

1. Project deaths and adaptation costs for each year to 2300.
 1. Multiply by the VSL.
 2. Discount at 2%.
2. Mortality-only social cost of carbon (Carleton et al.):
 - \$36.6 per ton of CO₂ under high emissions (RCP8.5), 2019 dollars.
 - 90% range: -\$7.8 to \$73.0.
 - \$17.1 per ton under moderate emissions (RCP4.5).
3. Lower under RCP4.5 because the response is convex: an extra ton does less damage when the world is cooler.
4. The models used for the 2010 SCC implied almost no mortality damage (FUND: 0.33 deaths per 100,000 in 2100).
5. → mortality ends up being one of the largest pieces of the current social cost of carbon.

Takeaways

1. Mortality responds to temperature the same way my electricity bill does: more/less U-shaped with a steep hot tail.
2. The curve is flatter where people are richer and used to heat. Adaptation is costly, though.
3. By 2100, climate change adds deaths on the scale of today's major causes, concentrated in hot, poor places.
4. The VSL converts risk reductions to dollars. Which VSL you use in poor countries changes the answer.
5. External costs of climate change on mortality alone is worth roughly \$20-40 per ton of CO₂.

What climate damages are we missing?

1. We've looked at electricity demand, agriculture, and heat-related mortality in depth?
 1. Most of these focused on temperature responses...
 2. What else is missing?
2. We've also looked at the temperature-GDP relationship.
 1. What's included in GDP? What's not?
3. The social cost of carbon adds up the monetized damages from each of the damage sectors (next class).

For next class...

1. Monday, 10/12: the social cost of carbon.
 - Take a look at the Rennert et al. article
2. Wednesday, 10/14: Exam 2 review. Bring questions.
3. Problem set 2 due Friday, 10/16, 11:59 PM.
4. www.caseyjwichman.com/econ4210#week8

See you on Monday!

caseyjwichman.com/econ4210/#week8